

Full Length Article

A hierarchical and privacy-preserving intrusion detection framework for SAGIN-enabled IIoT using graph neural networks and deep Q-learning

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ABSTRACT

Space–Air–Ground Integrated Networks (SAGIN) supporting Industrial Internet of Things (IIoT) applications impose stringent requirements on intrusion detection due to node mobility, heterogeneous link characteristics, constrained edge resources, and cross-tier attack propagation. Most standard security methods rely on one main system or some fixed settings that makes it hard to respond quickly when the network changes or when there is not much power. This paper presents HSP-SR, a hierarchical intrusion detection and response framework structured according to the physical segmentation of SAGIN. At the ground tier, a compact autoencoder performs event-driven anomaly detection using an adaptive threshold conditioned on residual energy, temporal variation, and local entropy, thereby restricting unnecessary uplink transmissions. The air tier applies an attention-based fusion model that assigns normalized weights to incoming alerts according to dynamically updated trust scores, entropy measures, and temporal freshness, producing a filtered aggregate representation. At the space tier, a time-variant graph neural network models the SAGIN topology as a weighted graph and infers cross-domain intrusion patterns through multi-hop message passing and neighborhood statistics. Confidentiality of inter-tier communication is maintained through elliptic curve key exchange combined with AES-based payload encryption, while trust values are updated using hash-chain-derived credentials and divergence-aware drift control. Adaptive mitigation decisions are determined through a Deep Q-Network formulated within a Markov Decision Process, where the reward function jointly accounts for threat magnitude, latency overhead, and energy expenditure. The CSE-CIC-IDS2018 dataset underwent experimental testing under six different attack scenarios which produced a 99.42% detection accuracy and 0.72% false positive rate and 22.1% lower energy usage than previous benchmarks while keeping detection times under 130 milliseconds for all test scenarios.

1. Introduction

The combination of non-terrestrial networking with learning-assisted communication control and cyber-physical integration systems creates new wide-area connectivity systems which use multiple domain infrastructure types (Chen et al., 2024b; Sefati & Halunga, 2023; Wang et al., 2023; Xiao et al., 2024). SAGINs achieve expanded land-based network reach through their combination of satellite and aerial components which create communication paths in areas with limited con-

nectivity and ensure network continuity during infrastructure outages (Awan et al., 2026; Buyuksar et al., 2024; Xu et al., 2023b). The deployment of Industrial Internet of Things (IIoT) systems brings about numerous sensors and actuators and controllers which need to operate under specific conditions that include restricted latency periods and reliable data transfer rates and continuous system accessibility (Liu et al., 2025; Wang et al., 2023). The SAGIN system operates with IIoT technology which allows remote industrial sites and mobile operations to function independently from standard coverage and deployment requirements

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(Awan et al., 2026; Shen et al., 2023). The system faces multiple complex control and security challenges because of its combined operation which includes handling different time synchronization requirements based on propagation delay variations and dealing with mobile network topology changes and spectrum usage conflicts between different network layers and ensuring secure system operations through links with different performance levels (Chen et al., 2025; Ding et al., 2025; Tu et al., 2024; Zhang et al., 2024a).

Security in SAGIN-enabled IIoT is constrained by both the attack surface created by cross-tier interconnection and the limited computational and energy budgets at the ground edge (Al-Quayyed et al., 2025; Awan et al., 2022; Zhang et al., 2025). The authentication systems and key-management systems become more susceptible to attacks because multi-hop routing and network disruptions occur together (Jin et al., 2025). The different network tier latency levels enable attackers to perform synchronized attacks because verification delays and security policy inconsistencies create system weaknesses (Cao et al., 2025; Zhang et al., 2024c). The research investigates two interconnected needs which combine distributed intrusion detection systems with cross-layer monitoring capabilities and maintain confidentiality for data exchanged between system tiers under both real-time and energy-efficient operating conditions (Awan et al., 2022; Poongodi & Hamdi, 2023). The target outcome involves creating a security design system which detects coordinated attacks while protecting sensitive data through protected data pathways under the conditions of UAV relay operations and satellite communication support (Awan et al., 2025, 2026).

Operational SAGIN-IIoT deployments require security functions that remain effective under mobility, intermittent connectivity, and constrained edge resources (Awan et al., 2025; Bakambekova et al., 2024; Sisi & Souiri, 2024). Trustworthy and low-latency security control is necessary when telemetry and command streams directly affect physical processes, including precision agriculture, autonomous logistics, and distributed energy infrastructure (Ding et al., 2025; Yang et al., 2026; Zhao et al., 2025). A substantial fraction of existing IIoT and SAGIN security designs assumes centralized inspection, continuous backhaul availability, or cryptographic stacks whose computation and signaling overhead is misaligned with resource-limited field deployments (Jin et al., 2025; Yang et al., 2026; Zhao et al., 2025). As a result, current architectures often fail to satisfy availability targets and autonomous operation requirements when network composition changes due to UAV mobility, tier handovers, or sensor churn (Awan et al., 2026; Xie et al., 2025; Zhang et al., 2024b; Zhou et al., 2023). This study targets a layer-aligned, energy-constrained security design that provides real-time intrusion detection, preserves confidentiality during inter-tier exchange, and adapts to topology variation without requiring persistent connectivity (Awan et al., 2025; Xie et al., 2025; Zhang et al., 2024b; Zhou et al., 2023).

Hierarchical Secure Privacy-SAGIN Real-time (HSP-SR) instantiates this objective through a three-tier detection and response design mapped to the SAGIN physical layers. At the ground tier, lightweight anomaly detectors execute event-triggered inference on constrained IIoT nodes using compact feature representations to reduce computation and uplink load. The air tier performs regional aggregation through an attention-based fusion model which applies weights to alerts based on source trust and uncertainty and temporal freshness to eliminate unimportant reports while maintaining important cross-node patterns. The space/control tier contains a graph-based intrusion reasoning module which uses time-variant graphs to show SAGIN connectivity and performs topology-aware inference to detect coordinated attacks that spread between different layers. The system maintains Inter-tier confidentiality through asymmetric session initialization which protects alert payloads through symmetric encryption with ephemeral keys for brief communication periods. The reinforcement learning agent chooses mitigation decisions through an optimization process which determines response actions based on threat and latency and energy constraints to minimize false alarms and minimize disruption during adaptive defense execution.

The principal technical contributions of this study are summarized as follows:

1. The SAGIN physical structure received a hierarchical intrusion detection system which followed its organizational layers. The ground layer detects local anomalies when resources become scarce but the air layer unites and purifies alert information and the space layer performs graph inference to identify attacks which move between different domains. The tiered decomposition system controls edge computing resources while maintaining complete situational awareness across the entire system.
2. The air tier operates with an alert fusion system which determines alert weights through the combination of trust values with uncertainty measures and temporal recency information. The system decreases unnecessary data transmission while it weakens untrustworthy signals which produces quantifiable better results for both detection precision and system operational speed.
3. A Deep Q-Network operates within a Markov Decision Process to create an adaptive mitigation framework. The policy determines its response actions through a reward function which combines threat severity assessment with latency overhead and energy consumption to make defensive choices based on current network conditions.

The structure of this article can be summarised as follows. [Section 2](#) provides an in-depth review of foundational technologies and recent developments relevant to SAGIN-enabled IIoT environments. The proposed multi-tier HSP-SR framework is introduced in [Section 3](#), detailing each component's functional logic and design considerations. [Section 4](#) outlines the simulation setup and key evaluation metrics. This is followed by a reflective discussion of observed trends and implications in [Section 5](#). Finally, [Section 6](#) concludes the article.

2. Background literature analysis

The SAGIN research organization has moved its security design approach from protecting individual domains to managing security across different operational levels because IIoT systems now use edge analytics and aerial relaying and satellite coordination. The first line of work focuses on resource-limited operation because security and networking functions need to function under combined energy and latency restrictions. The research community has worked to create energy-conscious coordination systems which optimize task distribution and data transfer performance through various SAGIN network connections (Liu et al., 2022a; Qin et al., 2024). Reinforcement-based policies were reported to improve allocation decisions across UAV relays and satellite nodes when compute budgets are bounded and channel quality varies over time (Liu et al., 2022a). Research on congestion and processing overhead implemented service function chaining and federated scheduling to reduce queuing delay and redundant computation by using topology-aware path embedding and task granularity alignment (Zhang et al., 2022a,b). The requirement for software-defined SAGINs to organize controllers in a hierarchical structure emerges because network signaling becomes unbalanced when nodes move which helps address problems that arise from changes in network topology structure (Chen et al., 2022).

The second research stream shows that trust establishment and authentication functions as critical obstacles for SAGIN-enabled IIoT because the unstable network links and slow data transfer rates prevent successful identity verification and secure key establishment. Blockchain-based constructions, including distributed ledgers and threshold signatures, have been used to support federated learning and verifiable identity exchange across sea-air-space domains (Chen et al., 2024a; Xie et al., 2026; Zhang et al., 2024c). The research presents two solutions which use session-oriented credentialing and satellite-terrestrial mutual authentication protocols to verify users through lossy networks at low processing costs for restricted nodes (Liu et al., 2022b). Other approaches reduce authentication overhead via

Table 1
Comparative analysis of existing approaches and the proposed HSP-SR framework.

Ref.	Core Focus and Technique	Identified Limitations
(Liu et al., 2022b), (Zhang et al., 2024c)	Trust establishment via authentication protocols and blockchain-based credential sharing	Lack of real-time adaptability; absence of fine-grained trust evolution under changing topology
(Xu et al., 2023a), (Li et al., 2025)	Federated anomaly detection with adaptive local learning	Communication cost reduction achieved but global attack patterns missed; weak topological awareness
(Zhang et al., 2022a), (Zhang et al., 2022b)	Service Function Chain (SFC)-based resource orchestration	No integrated security logic within task mapping; static task embedding models
(Wang et al., 2022), (Liu et al., 2022a)	Distributed offloading and cloud-edge coordination in SAGIN environments	Ignores role of coordinated intrusion response; no RL-based adaptation to attack states
(Chen et al., 2022), (Qin et al., 2024)	Controller hierarchy and resource allocation using multi-to-one matching	Omits integration with lightweight security primitives or detection under resource constraints
(Quy et al., 2024), (Deebak et al., 2022)	Quantum FL, searchable encryption for IIoT	Fails to address hierarchical inference, spatio-temporal alert correlation, or multi-tiered privacy logic
(Cai et al., 2024), (Chen et al., 2024a)	Secure scheduling and federated threshold signatures in heterogeneous domains	Limited attention to multi-level alert significance and trust-weighted decisions under load

quantum-safe primitives and multi-party consensus mechanisms tailored to space-ground mobility (Li et al., 2025; Quy et al., 2024). Privacy protection during coordination has also been studied through encrypted indexing, multi-keyword search, and secure scheduling logic, targeting leakage reduction when nodes exchange semantically rich metadata (Cai et al., 2024; Deebak et al., 2022).

The joint consideration of load balancing, privacy enforcement, and trust-aware detection remains an open systems problem in SAGIN-supported IIoT. Recent studies have combined federated anomaly detection with reinforcement-guided routing to reduce model divergence and signaling overhead in satellite-assisted deployments (Li et al., 2025; Xu et al., 2023a). Task-sensitive offloading strategies and delay-aware partitioning between cloud, edge, and device layers have further improved the reuse of locally trained security models under time-varying SAGIN connectivity (Wang et al., 2022). In parallel, privacy-constrained transfer learning has incorporated task obfuscation and differential masking to maintain inference accuracy while limiting information leakage in low-latency applications (Li et al., 2025). These approaches address specific components of distributed security, but typically evaluate performance under isolated objectives.

The current SAGIN-IIoT security systems operate with distinct subsystems which handle resource allocation and trust management and intrusion detection functions independently. The system divides its operations into separate sections which prevents it from making unified changes when its network structure and connection strength and security threats change at the same time. The previous security systems depend on single-layer detection logic and static credential updates and fixed mitigation rules which do not provide sufficient protection against mobility-driven graph changes and backhaul outages. The primary design obstacle requires the system to operate at low power consumption while it continues to identify security threats which occur between different system tiers based on time. The HSP-SR framework solves this problem through its method which links intrusion detection to physical tier organization and implements trust-based alert combination at the air layer and performs real-time trust management and protection choice optimization. A structured comparison with representative approaches is presented in Table 1.

3. Proposed HSP-SR framework

The HSP-SR framework solves problems which occur because of operational limitations and security needs that change frequently in SAGIN-enabled IIoT systems. The framework includes multiple security layers which defend distributed systems against attacks while it adapts to threats and maintains user privacy during coordination operations. The framework is structurally aligned with the physical layers of SAGIN. This section introduces its hierarchical design, explains layer-specific functionality, discusses inter-tier data handling, and outlines how threat

context is interpreted and acted upon under energy, latency, and trust-aware constraints.

3.1. System architecture overview

HSP-SR follows a vertically tiered design that matches the operational strata of a Space-Air-Ground Integrated Network. The stack is partitioned into three functionally separated planes (ground, air, space) to keep sensing, intermediate fusion, and global correlation within the compute, energy, and link budgets of each tier. Information exchange is restricted to structured alert messages and filtered telemetry, which limits uplink load and reduces exposure of raw traffic traces. At the ground tier, industrial Internet of Things endpoints and proximal edge nodes run lightweight anomaly detectors for streaming telemetry. As represented by Fig. 1, each node performs local feature extraction and online inference under constrained power and bandwidth. When the detection score crosses an adaptive decision threshold, the node emits a compact alert record containing metadata (e.g., time index, source identifier, anomaly score, and summary statistics) rather than raw payloads. Alerts are transmitted asynchronously to the air tier, and normal traffic continues without continuous reporting.

The air tier comprises unmanned aerial vehicles and high-altitude relays that provide regional aggregation and triage. These platforms execute an attention-based fusion module that combines alerts from multiple ground clusters. Fusion weights are computed from trust-conditioned reliability scores and context descriptors, which down-weights stale, low-confidence, or inconsistent reports. Only the fused, high-salience events and their confidence annotations are forwarded to the space tier, reducing redundant messages from correlated ground sources. The space tier includes satellite nodes and fixed control stations with sufficient capacity for topology-aware, cross-region correlation. Incoming fused alerts are embedded into a time-varying security graph whose nodes represent SAGIN entities and whose edges capture connectivity and interaction patterns. Graph-based inference produces network-level threat states, supports detection of coordinated or multi-stage activity, and generates reconfiguration directives (e.g., route isolation, access policy updates, or inspection escalation) that propagate downward through the same tiered control path.

3.2. Layer-wise intrusion detection system

This section details the hierarchical intrusion detection architecture distributed across the ground, air, and space tiers. Each layer performs targeted threat analysis using context-specific intelligence modules—ranging from lightweight local detectors at the edge, to attention-guided alert fusion in the air, and global topological reasoning through graph-based inference in the space domain. The procedural overview of the

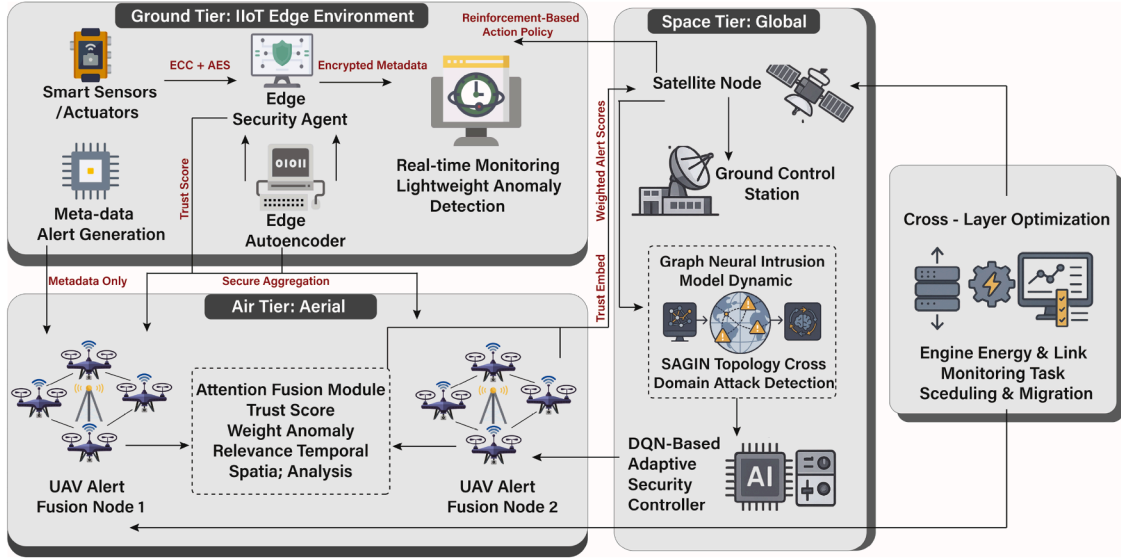


Fig. 1. Proposed HSP-SR architecture for secure SAGIN-IoT environments, consisting of three interlinked tiers—ground, air, and space—optimized through cross-layer reinforcement learning and trust-aware fusion. In the ground tier, encrypted metadata ($\mu_t^{(i)}, \zeta_t^{(i)}$) is generated via edge nodes using ECC-AES composite cryptography and transmitted to UAVs for secure aggregation (48). A lightweight anomaly detection module at the edge utilizes an autoencoder $\phi_{AE}(\cdot)$ with reconstruction loss $\mathcal{L}_{rec}$ to flag behavioral deviations. In the air tier, alert fusion is guided by attention weights $\alpha_i = \text{softmax}(e_i)$ over trust-weighted scores $\zeta_t^{(i)}$, forming an aggregate signal $\hat{S}_t$ via Eq. (51). Spatio-temporal relevance is captured using a function $\Psi(\mathcal{T}, \mathcal{L})$, aiding real-time filtering and prioritization. The space tier employs a dynamic graph neural network $\mathcal{G}(\mathcal{V}, \mathcal{E})$ to detect cross-domain threats, informed by SAGIN topology $\mathcal{T}_{sagin}$ and trust propagation matrices. A deep Q-network controller updates the action-value function $Q_\theta(S_t, a_t)$ per Eq. (71), adapting responses based on reward feedback $r_t = \Omega(D_t^{\text{threat}}, \mathcal{L}_t^{\text{lat}}, \mathcal{L}_t^{\text{cost}})$. The system continuously updates cross-layer parameters by minimizing a joint loss function $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DQN}} + \lambda_1 \mathcal{L}_{\text{priv}} + \lambda_2 \mathcal{L}_{\text{rec}}$.

space-tier graph-based intrusion reasoning pipeline is systematically encapsulated in Algorithm 1.

3.2.1. Ground tier: lightweight local anomaly detection

IIoT endpoints at the ground tier operate under tight energy and compute budgets; therefore, the local detector is implemented as an event-triggered autoencoding pipeline. Eqs. (1)–(8) formalize the processing stages from latent embedding to uplink-aware objective, with message generation conditioned on an adaptive residual test.

$$\mathcal{Z}_t = \sigma(\mathbf{W}_1 \cdot \phi(\mathbf{X}_t) + \mathbf{b}_1) \quad (1)$$

In Eq. (1), $\mathbf{X}_t \in \mathbb{R}^d$ denotes the d -dimensional sensory feature vector observed at discrete time index t . The mapping $\phi(\cdot)$ represents a lightweight feature projection (e.g., normalization, windowing, or compact basis transform) applied locally before encoding. The trainable matrix $\mathbf{W}_1$ and bias vector $\mathbf{b}_1$ parameterize the encoder, while $\sigma(\cdot)$ denotes a pointwise nonlinearity used to bound or reshape activations. The latent vector $\mathcal{Z}_t$ is the compressed representation used for downstream reconstruction and residual scoring.

$$\Delta_t = \|\mathbf{X}_t - \mathcal{G}(\mathcal{Z}_t)\|_2^2 \quad (2)$$

Eq. (2) defines the reconstruction residual Δ_t as the squared ℓ_2 norm of the difference between the original observation $\mathbf{X}_t$ and its reconstruction $\mathcal{G}(\mathcal{Z}_t)$, where $\mathcal{G}(\cdot)$ is a lightweight decoder. The operator $\|\cdot\|_2$ denotes the Euclidean norm, and the exponent 2 produces a nonnegative scalar deviation score used as the detection statistic at time t .

$$\mathbf{1}_{\text{alert}}^t = \begin{cases} 1, & \text{if } \Delta_t > \tau_g(\mathcal{E}_n, \mathcal{T}_t, \Lambda_t) \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

In Eq. (3), $\mathbf{1}_{\text{alert}}^t \in \{0, 1\}$ is a binary event flag indicating whether the node emits an alert at time t . The decision compares the residual Δ_t against a threshold $\tau_g(\cdot)$ that adapts to node state and local uncertainty. The term $\mathcal{E}_n$ denotes the residual energy budget of node n (e.g., Joules or normalized units), $\mathcal{T}_t$ denotes a temporal factor at time t (e.g., elapsed time since last report, window index, or duty-cycle phase), and Λ_t denotes an uncertainty descriptor derived from local observations, defined

Algorithm 1: Space tier: graph-based intrusion reasoning in SAGIN.

Input: Graph snapshot $\mathcal{G}_t$ with node features $\mathbf{X}_t$ and adjacency matrix $\mathbf{A}_t$
Output: Intrusion class prediction $\hat{y}_v$ and anomaly map $\mathcal{M}_t$

- 1 **Initialize:** Construct graph $\mathcal{G}_t$ as per Eq. (27); extract node features $\mathbf{X}_t$ from Eq. (28); form adjacency matrix using Eq. (29)
- 2 **foreach** node $v \in \mathcal{V}_t$ **do**
- 3 Set initial embedding $\mathbf{H}_v^{(0)}$ using Eq. (31);
- 4 **for** $l = 1$ to L **do**
- 5 **foreach** node $v \in \mathcal{V}_t$ **do**
- 6 **foreach** neighbor $u \in \mathcal{N}(v)$ **do**
- 7 Compute spatial term δ_{vu} using Eq. (36);
- 8 Form edge embedding e_{vu} via Eq. (35);
- 9 Evaluate attention weight $\alpha_{vu}^{(l)}$ using Eq. (34);
- 10 Update node embedding $\mathbf{H}_v^{(l)}$ as per Eq. (32);
- 11 **foreach** node $v \in \mathcal{V}_t$ **do**
- 12 Aggregate multi-layer embeddings to obtain $\mathbf{Z}_v$ using Eq. (33);
- 13 Compute neighbor mean μ_v using Eq. (37);
- 14 Compute variance σ_v^2 from Eq. (38);
- 15 Derive intrusion score $\mathcal{R}_v$ using Eq. (39);
- 16 Classify via softmax to obtain y_v as per Eq. (40);
- 17 Predict class label $\hat{y}_v$ using Eq. (41);
- 18 Compute total loss $\mathcal{L}_{\text{GNN}}$ via Eq. (42);
- 19 Update model parameters θ_g using Eq. (43);
- 20 **Construct Anomaly Map:**
- 21 **foreach** node $v \in \mathcal{V}_t$ **do**
- 22 Add $(v, \hat{y}_v, \mathcal{R}_v)$ to $\mathcal{M}_t$ using Eq. (44);
- 23 Cluster $\mathcal{M}_t$ spatio-temporally to form $\mathcal{C}_t$ using Eq. (45);
- 24 **return** $\hat{y}_v, \mathcal{M}_t, \mathcal{C}_t$

explicitly below.

$$\tau_g(\mathcal{E}_n, \mathcal{T}_t, \Lambda_t) = \alpha \cdot \mathcal{E}_n^{-1/2} + \beta \cdot \ln(1 + \mathcal{T}_t^2) + \gamma \cdot \psi(\Lambda_t) \quad (4)$$

Eq. (4) specifies the adaptive threshold $\tau_g(\cdot)$ as a weighted sum of three terms. The coefficient α scales the energy-sensitive component $\mathcal{E}_n^{-1/2}$, which increases the threshold as available energy decreases. The coefficient β scales the temporal term $\ln(1 + \mathcal{T}_t^2)$, where $\ln(\cdot)$ is the natural logarithm and $\mathcal{T}_t^2$ captures the magnitude of the time factor. The coefficient γ scales the uncertainty term $\psi(\Lambda_t)$, where $\psi(\cdot)$ maps the distributional descriptor Λ_t to a scalar uncertainty penalty.

$$\psi(\Lambda_t) = -\sum_{i=1}^K \lambda_i^t \cdot \log(\lambda_i^t + \varepsilon) \quad (5)$$

In Eq. (5), $\Lambda_t = \{\lambda_1^t, \dots, \lambda_K^t\}$ denotes a categorical posterior distribution over K local observation types or states at time t , where each $\lambda_i^t \in [0, 1]$ and $\sum_{i=1}^K \lambda_i^t = 1$. The summation index i ranges over the K categories. The function $\log(\cdot)$ denotes the natural logarithm, and $\varepsilon > 0$ is a small constant that prevents undefined values when λ_i^t approaches zero. The scalar $\psi(\Lambda_t)$ is an entropy-like measure that increases with distributional dispersion, thereby adjusting τ_g in Eq. (4) according to local uncertainty. A triggered decision in Eq. (3) gets message formation and energy-aware prioritization, as defined next.

$$\mathbf{M}_t = \mathbb{I}_{\text{alert}=1} \cdot (\mathcal{H}_t \odot \eta_t) \quad (6)$$

Eq. (6) defines the transmitted packet $\mathbf{M}_t$ constructed at time t . The indicator $\mathbb{I}_{\text{alert}=1}$ equals 1 when the alert flag $\mathbf{1}_{\text{alert}}^t$ is active and equals 0 otherwise, ensuring that $\mathbf{M}_t = \mathbf{0}$ when no event is raised. The vector $\mathcal{H}_t$ denotes the encoded alert signature (e.g., compressed features, residual statistics, and identifiers) produced locally upon triggering. The operator $\odot$ denotes the Hadamard (element-wise) product. The scalar or element-wise weight η_t modulates the payload magnitude according to the node energy state, thereby controlling the effective contribution of the packet under constrained resources.

$$\eta_t = \exp\left(-\mu \cdot \frac{P_t^{\text{comp}} + P_t^{\text{comm}}}{\mathcal{E}_n}\right) \quad (7)$$

In Eq. (7), η_t is defined via an exponential decay, where $\exp(\cdot)$ is the natural exponential function. The terms P_t^{comp} and P_t^{comm} denote the energy consumed at time t by computation (local inference and feature processing) and communication (radio transmission), respectively. The denominator $\mathcal{E}_n$ is the residual energy of node n , consistent with Eq. (3). The scalar $\mu > 0$ controls the decay rate; larger μ produces faster suppression of packet weight as the ratio $(P_t^{\text{comp}} + P_t^{\text{comm}})/\mathcal{E}_n$ increases.

$$\min_{\mathcal{F}_g} \mathbb{E}\left[\frac{\|\mathbf{M}_t\|_1}{\|\mathcal{H}_t\|_1}\right] + \lambda \cdot \mathbb{E}[\eta_t^{-1}] \quad (8)$$

Eq. (8) defines the ground-tier objective that selects a transmission policy $\mathcal{F}_g$ (e.g., feature retention, sparsification level, or message formatting rule) to control uplink cost. The operator $\mathbb{E}[\cdot]$ denotes expectation over time and operating conditions (e.g., traffic regimes, node states, and observation distributions). The ℓ_1 norm $\|\cdot\|_1$ sums absolute element magnitudes; thus, $\|\mathbf{M}_t\|_1/\|\mathcal{H}_t\|_1$ quantifies the proportion of transmitted content relative to the full alert signature. The scalar $\lambda \geq 0$ weights the energy-related penalty term, and η_t^{-1} penalizes low-energy operation because η_t decreases with rising energy expenditure in Eq. (7). Minimizing the combined objective reduces transmitted mass while discouraging policies that operate near depleted energy states.

3.2.2. Air tier: attention-based alert fusion

The air tier aggregates alert messages received from N_t active ground devices at time index t and computes a trust-conditioned fusion vector. Let $\mathbf{M}_t^{(i)} \in \mathbb{R}^d$ denote the alert vector transmitted by device i . The attention projection, trust modulation, normalized weighting, and initial

fusion are defined in Eqs. 9.

$$\mathbf{A}_t^{(i)} = \varphi(\mathbf{M}_t^{(i)} \mathbf{W}_q + \mathbf{b}_q), \quad (9)$$

$$\Theta^{(i)} = \omega_1 \mathcal{T}_i + \omega_2 \kappa_i + \omega_3 \rho_i, \quad (10)$$

$$\alpha_t^{(i)} = \frac{\exp(\Theta^{(i)} \mathbf{A}_t^{(i)})}{\sum_{j=1}^{N_t} \exp(\Theta^{(j)} \mathbf{A}_t^{(j)})}, \quad (11)$$

$$\mathbf{F}_t = \sum_{i=1}^{N_t} \alpha_t^{(i)} \mathbf{M}_t^{(i)} \quad (12)$$

In Eq. (9), the vector $\mathbf{A}_t^{(i)}$ is obtained by linearly projecting $\mathbf{M}_t^{(i)}$ using learnable parameters $\mathbf{W}_q \in \mathbb{R}^{d \times d_q}$ and bias $\mathbf{b}_q \in \mathbb{R}^{d_q}$, followed by a non-linear activation $\varphi(\cdot)$ such as ReLU or GELU. The resulting $\mathbf{A}_t^{(i)} \in \mathbb{R}^{d_q}$ is the attention-space embedding for device i . The scalar $\Theta^{(i)}$ combines three trust-related components. The term $\mathcal{T}_i$ denotes a temporal decay factor associated with device i , κ_i represents a mobility volatility index, and ρ_i denotes an entropy-based uncertainty measure. The coefficients ω_1 , ω_2 , and ω_3 are nonnegative scalars that determine the relative contribution of these components.

The weight $\alpha_t^{(i)}$ is a normalized compatibility coefficient computed via a softmax over all N_t devices. The numerator evaluates the exponential of the inner product between $\Theta^{(i)}$ and $\mathbf{A}_t^{(i)}$, and the denominator enforces $\sum_{i=1}^{N_t} \alpha_t^{(i)} = 1$. The fused vector $\mathbf{F}_t \in \mathbb{R}^d$ is then obtained as the weighted linear combination of the original alert vectors. The fused representation is subsequently refined through a gating and filtering stage defined in Eq. (13).

$$\mathcal{G}_t = \sigma(\mathbf{F}_t \mathbf{W}_g + \mathbf{b}_g), \quad (13)$$

$$S_t^{(i)} = \Gamma(\mathbf{A}_t^{(i)}, \mathbf{M}_t^{(i)}, \Theta^{(i)}), \quad (14)$$

$$\hat{\alpha}_t^{(i)} = \mathbb{I}_{S_t^{(i)} > \delta} \alpha_t^{(i)}, \quad (15)$$

$$\hat{\mathbf{F}}_t = \sum_{i=1}^{N_t} \hat{\alpha}_t^{(i)} \mathcal{G}_t \quad (16)$$

In Eq. (13), $\mathcal{G}_t$ is a gated transformation of $\mathbf{F}_t$ using learnable parameters $\mathbf{W}_g$ and $\mathbf{b}_g$, with $\sigma(\cdot)$ denoting a sigmoid activation that maps each element to $(0, 1)$. The scalar $S_t^{(i)}$ is computed by a scoring function $\Gamma(\cdot)$ that takes as input the embedding $\mathbf{A}_t^{(i)}$, the raw alert $\mathbf{M}_t^{(i)}$, and the trust coefficient $\Theta^{(i)}$. The threshold δ is a predefined scalar. The indicator function $\mathbb{I}_{S_t^{(i)} > \delta}$ equals 1 if the condition is satisfied and 0 otherwise, thereby suppressing contributions from low-scoring devices. The filtered fusion vector $\hat{\mathbf{F}}_t$ aggregates only retained components. The trust-related components are defined explicitly in Eqs. (17).

$$\rho_i = -\sum_{k=1}^K p_k^{(i)} \log(p_k^{(i)} + \varepsilon), \quad (17)$$

$$\kappa_i = \frac{1}{T} \sum_{t'=t-T+1}^t \|\mathbf{l}_{t'}^{(i)} - \mathbf{l}_{t'-1}^{(i)}\|_2^2, \quad (18)$$

$$\mathcal{T}_i = \gamma \exp(-\tau(t - t_i^{\text{last}})) \quad (19)$$

In Eq. (17), ρ_i is the Shannon entropy of the posterior distribution $\{p_k^{(i)}\}_{k=1}^K$, where K denotes the number of alert categories and $\varepsilon > 0$ ensures numerical stability. The mobility index κ_i computes the average squared displacement over a sliding window of length T , where $\mathbf{l}_{t'}^{(i)} \in \mathbb{R}^m$ is the spatial coordinate of device i at time t' and $\|\cdot\|_2$ denotes the Euclidean norm. The temporal factor $\mathcal{T}_i$ decays exponentially with elapsed time since the last activity t_i^{last} , with γ and τ controlling magnitude and decay rate. The filtered vector is normalized and converted to a priority representation as defined in Eq. (20).

$$\tilde{\mathbf{F}}_t = \frac{\hat{\mathbf{F}}_t}{\|\hat{\mathbf{F}}_t\|_2 + \xi}, \quad (20)$$

$$P_t = \mathcal{M}(\tilde{\mathbf{F}}_t, \{\hat{\alpha}_t^{(i)}\}_{i=1}^{N_t}) \quad (21)$$

In Eq. (20), $\tilde{\mathbf{F}}_t$ is obtained by dividing $\hat{\mathbf{F}}_t$ by its ℓ_2 norm plus a small constant $\xi > 0$ to avoid division by zero. The function $\mathcal{M}(\cdot)$ maps the normalized vector and retained weights to a priority map $\mathcal{P}_t$, which encodes alert relevance and weight concentration. Model training uses the objective defined in Eq. (22).

$$\mathcal{E}_t^{\text{att}} = \sum_{i=1}^{N_t} \alpha_i^{(t)} \rho_i, \quad (22)$$

$$\Delta_\alpha = \sum_{i=1}^{N_t} \left| \alpha_i^{(t)} - \frac{1}{N_t} \right|, \quad (23)$$

$$\mathcal{L}_{\text{fusion}} = \lambda_1 \mathcal{E}_t^{\text{att}} + \lambda_2 \Delta_\alpha + \lambda_3 \|\tilde{\mathbf{F}}_t - \mathbf{y}_t\|_2^2 \quad (24)$$

In Eq. (22), $\mathcal{E}_t^{\text{att}}$ is the entropy-weighted attention term, Δ_α penalizes deviation from uniform weighting, and $\mathbf{y}_t$ denotes the ground-truth supervision vector at time t . The scalars λ_1 , λ_2 , and λ_3 regulate the relative contribution of entropy regularization, diversity control, and reconstruction fidelity. The ℓ_2 norm quantifies squared deviation from $\mathbf{y}_t$. Parameter updates follow gradient descent:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta_t} \mathcal{L}_{\text{fusion}} \quad (25)$$

In Eq. (25), θ_t denotes the vector of all learnable parameters at iteration t , $\eta > 0$ is the learning rate, and $\nabla_{\theta_t} \mathcal{L}_{\text{fusion}}$ is the gradient of the loss with respect to θ_t . Finally, the scalar reliability score is computed as

$$\zeta_t = \frac{1}{1 + \exp(-\mathbf{v}^\top \tilde{\mathbf{F}}_t)} \quad (26)$$

In Eq. (26), $\mathbf{v}$ is a learnable projection vector, $(\cdot)^\top$ denotes transpose, and $\zeta_t \in (0, 1)$ is the sigmoid-transformed confidence associated with the fused alert at time t .

3.2.3. Space tier: graph-based global intrusion reasoning

At the space tier, satellite control nodes and orbital processing units integrate aggregated alerts into a time-indexed graph representation of the SAGIN infrastructure. Intrusion activity is modeled through heterogeneous graph embeddings and multi-hop message passing with edge-level attention, enabling detection of cross-tier and temporally correlated attack propagation.

$$\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathcal{A}_t) \quad (27)$$

$$\mathbf{X}_t = [\mathbf{x}_1^t, \dots, \mathbf{x}_{|\mathcal{V}_t|}^t] \quad (28)$$

$$\mathbf{A}_t = [a_{ij}^t] \in \mathbb{R}^{|\mathcal{V}_t| \times |\mathcal{V}_t|} \quad (29)$$

$$\mathcal{F}_{\text{GNN}}^t = f(\mathcal{G}_t, \mathbf{X}_t; \theta_g) \quad (30)$$

Eq. (27) defines the temporal graph $\mathcal{G}_t$, where $\mathcal{V}_t$ is the set of nodes (ground stations, aerial relays, satellites), $\mathcal{E}_t$ is the set of communication links, and $\mathcal{A}_t$ represents structural connectivity. Eq. (28) specifies the node feature matrix $\mathbf{X}_t$, in which each $\mathbf{x}_i^t$ encodes operational indicators such as energy state, link statistics, or aggregated threat descriptors for node i . Eq. (29) defines the adjacency matrix $\mathbf{A}_t$, where a_{ij}^t denotes the connection weight between nodes i and j . Eq. (30) expresses the graph neural mapping $\mathcal{F}_{\text{GNN}}^t$ parameterized by θ_g , which processes structural and attribute information jointly. Node embeddings are computed through layer-wise propagation as defined in Eqs. (31)–(33).

$$\mathbf{H}_v^{(0)} = \mathbf{x}_v^t \quad (31)$$

$$\mathbf{H}_v^{(l)} = \sigma \left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(l)} \mathbf{W}_u^{(l)} \mathbf{H}_u^{(l-1)} + \mathbf{b}^{(l)} \right) \quad (32)$$

$$\mathbf{Z}_v = \text{Agg}_{l=1}^L \mathbf{H}_v^{(l)} \quad (33)$$

Eq. (31) initializes the embedding of node v using its input feature vector $\mathbf{x}_v^t$. Eq. (32) updates the embedding at layer l by aggregating neighbor embeddings $\mathbf{H}_u^{(l-1)}$ for $u \in \mathcal{N}(v)$, where $\mathcal{N}(v)$ denotes the neighborhood of v . The matrix $\mathbf{W}^{(l)}$ and bias $\mathbf{b}^{(l)}$ are layer-specific parameters, $\alpha_{vu}^{(l)}$ is the attention coefficient for edge (v, u) , and $\sigma(\cdot)$ is a nonlinear activation. Eq. (33) aggregates representations across L layers to produce

the final embedding $\mathbf{Z}_v$. The attention coefficients are computed using edge-conditioned scoring as defined in Eqs. (34)–(35).

$$\alpha_{vu}^{(l)} = \frac{\exp(\phi(\mathbf{H}_v^{(l-1)}, \mathbf{H}_u^{(l-1)}, e_{vu}))}{\sum_{k \in \mathcal{N}(v)} \exp(\phi(\mathbf{H}_v^{(l-1)}, \mathbf{H}_k^{(l-1)}, e_{vk}))} \quad (34)$$

$$e_{vu} = \psi(\text{concat}(\mathbf{H}_v^{(l-1)}, \mathbf{H}_u^{(l-1)}, \delta_{vu})) \quad (35)$$

Eq. (34) defines $\alpha_{vu}^{(l)}$ as a softmax-normalized attention weight over neighbors of v , where $\phi(\cdot)$ is a compatibility function and $\exp(\cdot)$ is the exponential function. Eq. (35) computes the edge representation e_{vu} using a neural mapping $\psi(\cdot)$ applied to the concatenation of node embeddings and a spatial descriptor δ_{vu} . Spatial and neighborhood statistics are defined in Eqs. (36)–(38).

$$\delta_{vu} = \|\mathbf{p}_v - \mathbf{p}_u\|_2 + \xi \mathbb{I}_{[\text{tier}(v) \neq \text{tier}(u)]} \quad (36)$$

$$\mu_v = \text{mean}(\{\mathbf{Z}_u \mid u \in \mathcal{N}(v)\}) \quad (37)$$

$$\sigma_v^2 = \text{var}(\{\mathbf{Z}_u \mid u \in \mathcal{N}(v)\}) \quad (38)$$

Eq. (36) defines δ_{vu} as the Euclidean distance between positions $\mathbf{p}_v$ and $\mathbf{p}_u$, augmented by an inter-tier penalty ξ when nodes belong to different tiers. Eq. (37) computes the neighborhood mean embedding μ_v , and Eq. (38) computes the corresponding variance σ_v^2 . Node-level intrusion inference is defined in Eqs. (39)–(41).

$$\mathcal{R}_v = \zeta(\mu_v, \sigma_v^2, \mathbf{Z}_v) \quad (39)$$

$$\mathbf{y}_v = \text{softmax}(\mathbf{W}_o \mathcal{R}_v + \mathbf{b}_o) \quad (40)$$

$$\hat{y}_v = \arg \max_k [\mathbf{y}_v]_k \quad (41)$$

Eq. (39) computes a node-specific risk score $\mathcal{R}_v$ using the embedding $\mathbf{Z}_v$ and neighborhood statistics. Eq. (40) projects $\mathcal{R}_v$ using parameters $\mathbf{W}_o$ and $\mathbf{b}_o$, followed by softmax normalization to obtain class probabilities $\mathbf{y}_v$. Eq. (41) selects the predicted label $\hat{y}_v$ via the maximum probability index. Training is performed using the loss and update rules in Eqs. (42), (43).

$$\mathcal{L}_{\text{GNN}} = - \sum_{v \in \mathcal{V}_t} \sum_{k=1}^K y_v^{(k)} \log([\mathbf{y}_v]_k) \quad (42)$$

$$\theta_g \leftarrow \theta_g - \eta \nabla_{\theta_g} \mathcal{L}_{\text{GNN}} \quad (43)$$

Eq. (42) defines the cross-entropy loss between predicted probabilities $\mathbf{y}_v$ and one-hot labels $y_v^{(k)}$. Eq. (43) updates the parameter set θ_g using gradient descent with learning rate η . Finally, global anomaly mapping and clustering are defined in Eqs. (44)–(45).

$$\mathcal{M}_t = \{(v, \hat{y}_v, \mathcal{R}_v) \mid v \in \mathcal{V}_t\} \quad (44)$$

$$\mathcal{C}_t = \text{cluster}(\mathcal{M}_t, \text{spatial}, \text{temporal}) \quad (45)$$

Eq. (44) constructs a node-wise mapping of predicted class $\hat{y}_v$ and associated score $\mathcal{R}_v$. Eq. (45) applies clustering over spatial and temporal attributes to identify persistent or coordinated intrusion structures across the network.

3.3. Privacy-preserving data communication

To ensure confidential yet efficient data relay across heterogeneous SAGIN layers, the proposed model deploys a tier-aware encryption protocol where metadata is adaptively encrypted based on both data sensitivity and node privilege. Without disclosing raw observations, elliptic curve-based mutual key negotiation and lightweight symmetric encryption enable selective secure aggregation, particularly at the air tier, where full decryption is intentionally circumvented. The operational design of secure and privacy-adaptive communication is formalized in Algorithm 2.

$$\mathcal{E}_{\text{priv}}^{(i)} : y^2 \equiv x^3 + a_i x + b_i \pmod{\mathbb{P}_i}, \quad (46)$$

$$\mathcal{K}_i = \mathcal{H}(S_i \cdot \mathcal{P}_i), \quad (47)$$

Algorithm 2: Tier-aware privacy-preserving communication protocol.

```

1 for each node  $i$  in active session do
2   Construct elliptic curve  $\mathcal{E}_{\text{priv}}^{(i)}$  as per Eq. (46)
3   Generate shared key  $\mathcal{K}_i$  using Eq. (47)
4   Derive symmetric key  $\mathcal{K}_i^{\text{sym}}$  via Eq. (52)
5   Encrypt metadata  $\mu_t^{(i)} \parallel \zeta_t^{(i)}$  to form  $C_{ij}$  using Eq. (48)
6   Secure payload  $\mathbf{M}_t^{(i)}$  as  $\widehat{\mathcal{M}}_t^{(i)}$  using Eq. (49)
7 end
8 if UAV aggregator is active then
9   Aggregate all  $C_{ij}$  into  $\Pi_{\text{sum}}$  using Eq. (50)
10  Decode approximate alert state  $\widehat{S}_t$  via Eq. (51)
11  Validate synthesis using Eq. (53)
12 end
13 Compute leakage metric  $\Lambda_{\text{priv}}$  using Eq. (54)
14 Evaluate trade-off utility  $\Psi_t$  via Eq. (55)
15 Update privacy-preserving loss  $\mathcal{L}_{\text{priv}}$  by Eq. (56)

```

$$C_{ij} = \text{Enc}_{\text{ECC}}\left(\mu_t^{(i)} \parallel \zeta_t^{(i)}; \mathcal{K}_i\right), \quad (48)$$

$$\widehat{\mathcal{M}}_t^{(i)} = \text{AES}_{128}\left(\mathbf{M}_t^{(i)}, \mathcal{K}_i^{\text{sym}}\right) \quad (49)$$

Eq. (46) specifies the node-specific elliptic curve $\mathcal{E}_{\text{priv}}^{(i)}$ defined over the prime field $\mathbb{P}_i$, with curve parameters a_i and b_i assigned to node i . Eq. (47) defines the shared secret $\mathcal{K}_i$ by applying the hash function $\mathcal{H}(\cdot)$ to the scalar-point multiplication of the private scalar S_i and the peer public point P_j associated with node j . Eq. (48) encrypts the metadata tuple $\mu_t^{(i)} \parallel \zeta_t^{(i)}$, where $\mu_t^{(i)}$ is the metadata value produced by node i at time t and $\zeta_t^{(i)}$ is the associated confidence or trust indicator, using elliptic-curve encryption $\text{Enc}_{\text{ECC}}(\cdot)$ under key $\mathcal{K}_i$. Eq. (49) encrypts the alert payload $\mathbf{M}_t^{(i)}$ using the $\text{AES}_{128}(\cdot)$ block cipher with the symmetric key $\mathcal{K}_i^{\text{sym}}$ derived from $\mathcal{K}_i$. To support aggregation and validation at the air tier, the encrypted metadata are combined and decoded as follows.

$$\Pi_{\text{sum}} = \bigoplus_{i=1}^{N_t} C_{ij}, \quad (50)$$

$$\widehat{S}_t = \Theta_{\text{UAV}}^{-1}\left(\Pi_{\text{sum}}, \{\zeta_t^{(i)}\}_{i=1}^{N_t}\right), \quad (51)$$

$$\mathcal{K}_i^{\text{sym}} = \mathcal{F}_{\sigma}(\mathcal{K}_i, \rho_i, \mathcal{T}_i), \quad (52)$$

$$\Upsilon_t = \mathbb{V}\left(\widehat{S}_t, \bigcup_{i=1}^{N_t} \zeta_t^{(i)} \cdot \mathbf{h}_i\right) \quad (53)$$

Eq. (50) defines ciphertext aggregation Π_{sum} using the XOR operator $\bigoplus$ over the N_t ciphertexts C_{ij} available at time t . Eq. (51) defines the synthesized alert signal $\widehat{S}_t$ recovered by the UAV-side inverse decoder $\Theta_{\text{UAV}}^{-1}(\cdot)$ using the aggregated ciphertext Π_{sum} and the set of confidence indicators $\{\zeta_t^{(i)}\}_{i=1}^{N_t}$. Eq. (52) defines the symmetric key $\mathcal{K}_i^{\text{sym}}$ generated by the masking function $\mathcal{F}_{\sigma}(\cdot)$ from the shared secret $\mathcal{K}_i$, the uncertainty score ρ_i , and the temporal decay factor $\mathcal{T}_i$. Eq. (53) defines the verification output Υ_t computed by the verifier $\mathbb{V}(\cdot)$ from the synthesized signal $\widehat{S}_t$ and the aggregated trust-weighted signature set $\bigcup_{i=1}^{N_t} \zeta_t^{(i)} \cdot \mathbf{h}_i$, where $\mathbf{h}_i$ denotes the signature or hash-derived authentication vector associated with node i . The privacy and utility terms are quantified through entropy and divergence criteria, as follows.

$$\Lambda_{\text{priv}} = \frac{1}{N_t} \sum_{i=1}^{N_t} \left(\mathbb{H}(\mathbf{M}_t^{(i)}) - \mathbb{H}(\widehat{\mathcal{M}}_t^{(i)}) \right), \quad (54)$$

$$\Psi_t = \omega_1 \cdot \Lambda_{\text{priv}} - \omega_2 \cdot \mathcal{D}_{\text{KL}}(\widehat{S}_t \parallel \mathbf{y}_t), \quad (55)$$

$$\mathcal{L}_{\text{priv}} = \lambda_1 \cdot \Lambda_{\text{priv}} + \lambda_2 \cdot \|\mathbf{y}_t - \widehat{S}_t\|^2 + \lambda_3 \cdot \mathcal{R}(\mathcal{K}_i) \quad (56)$$

Eq. (54) defines the leakage measure Λ_{priv} as the average entropy gap across nodes, where $\mathbb{H}(\cdot)$ denotes Shannon entropy, $\mathbf{M}_t^{(i)}$ is the plaintext payload, and $\widehat{\mathcal{M}}_t^{(i)}$ is the corresponding ciphertext under $\text{AES}_{128}(\cdot)$. Eq. (55) defines the privacy-utility score Ψ_t using weights ω_1 and ω_2 , where $\mathcal{D}_{\text{KL}}(\cdot \parallel \cdot)$ is the Kullback-Leibler divergence between the synthesized signal $\widehat{S}_t$ and the reference label distribution $\mathbf{y}_t$. Eq. (56) defines the privacy-aware objective $\mathcal{L}_{\text{priv}}$, where λ_1 , λ_2 , and λ_3 weight the leakage term, the squared error $\|\mathbf{y}_t - \widehat{S}_t\|^2$, and the key regularization $\mathcal{R}(\mathcal{K}_i)$, respectively.

3.4. Trust and authentication scheme

To preserve transaction integrity in Space-Air-Ground Integrated Network environments with adversarial participants and non-uniform trust assumptions, this section specifies a trust and authentication scheme based on ephemeral credentials and hashchain validation. The resulting trust variables are injected into the detection and scheduling logic so that thresholding and verification effort depend on measured authentication strength and observed behavioral consistency.

$$\mathcal{H}_k^{(i)} = \mathcal{F}_{\text{hash}}\left(\mathcal{H}_{k-1}^{(i)} \parallel \mathbf{ID}_i \parallel \tau_k\right), \quad (57)$$

$$C_i^{(t)} = \zeta \cdot \mathcal{H}_k^{(i)} + (1 - \zeta) \cdot \mathcal{R}_i^{(t-1)}, \quad (58)$$

$$\mathcal{T}_i^{(t)} = \Upsilon\left(C_i^{(t)}, \eta_i^{(t)}, \xi_i^{(t)}\right), \quad (59)$$

$$\mathbb{T}_{\text{link}}^{i \leftrightarrow j} = \text{sigm}\left(\Theta_i^{\text{auth}} \cdot \Phi_j^{\text{cons}} - \delta \cdot \Gamma_{ij}^{\text{lat}}\right) \quad (60)$$

Eq. (57) defines the one-way hashchain state $\mathcal{H}_k^{(i)}$ for node i at index k , where $\mathcal{F}_{\text{hash}}(\cdot)$ is a cryptographic hash, $\mathcal{H}_{k-1}^{(i)}$ is the previous state, $\mathbf{ID}_i$ is the static node identifier, τ_k is a time nonce, and $\parallel$ denotes concatenation. Eq. (58) defines the time- t ephemeral credential $C_i^{(t)}$ as a convex combination of the current hash value $\mathcal{H}_k^{(i)}$ and the prior reputation score $\mathcal{R}_i^{(t-1)}$, where $\zeta \in [0, 1]$ sets the relative weight. Eq. (59) computes the trust score $\mathcal{T}_i^{(t)}$ via the composition function $\Upsilon(\cdot)$ from the credential $C_i^{(t)}$, the anomaly inertia term $\eta_i^{(t)}$, and the interaction freshness term $\xi_i^{(t)}$. Eq. (60) defines the bidirectional link trust $\mathbb{T}_{\text{link}}^{i \leftrightarrow j}$ between nodes i and j , where $\text{sigm}(\cdot)$ is the logistic sigmoid, Θ_i^{auth} denotes the authentication strength associated with node i , Φ_j^{cons} denotes the consistency score of node j , Γ_{ij}^{lat} is the latency penalty for the i - j link, and $\delta \geq 0$ scales the latency term.

$$\tau_{\text{dyn}}^{(i)} = \alpha_1 \cdot (1 - \mathcal{T}_i^{(t)}) + \alpha_2 \cdot \ln(1 + \rho_i), \quad (61)$$

$$P_{\text{verify}}^{(t)} = 1 - \exp\left(-\beta \cdot \mathcal{T}_i^{(t)} \cdot \mathbb{I}_{\text{alert}}^t\right), \quad (62)$$

$$\Delta \mathcal{T}_i = \lambda_1 \cdot \mathcal{D}_{\text{JSD}}(C_i^{(t)}, C_i^{(t-1)}) - \lambda_2 \cdot \Omega_i^{\text{mis}}, \quad (63)$$

$$\mathcal{T}_i^{(t+1)} = \text{clip}\left(\mathcal{T}_i^{(t)} - \Delta \mathcal{T}_i, 0, 1\right) \quad (64)$$

Eq. (61) defines the trust-conditioned detection threshold $\tau_{\text{dyn}}^{(i)}$ for node i , where α_1 and α_2 are scaling coefficients, $(1 - \mathcal{T}_i^{(t)})$ increases the threshold as trust decreases, ρ_i is an uncertainty index for node i , and $\ln(\cdot)$ is the natural logarithm. Eq. (62) defines the verification probability $P_{\text{verify}}^{(t)}$, where $\exp(\cdot)$ is the exponential function, $\beta > 0$ controls verification sensitivity, $\mathcal{T}_i^{(t)}$ is the current trust score, and $\mathbb{I}_{\text{alert}}^t \in \{0, 1\}$ indicates whether an alert is active at time t . Eq. (63) defines the trust drift $\Delta \mathcal{T}_i$, where λ_1 and λ_2 weight the Jensen-Shannon divergence $\mathcal{D}_{\text{JSD}}(\cdot, \cdot)$ between successive credentials and the misbehavior penalty Ω_i^{mis} , respectively. Eq. (64) updates the trust score by subtracting $\Delta \mathcal{T}_i$ from $\mathcal{T}_i^{(t)}$ and applying $\text{clip}(\cdot, 0, 1)$ to enforce the bounded range $[0, 1]$.

$$\mathbf{T}_{\text{global}} = \left[\mathcal{T}_i^{(t)} \cdot \mathbb{T}_{\text{link}}^{i \leftrightarrow j} \right]_{i,j \in \mathcal{V}_t}, \quad (65)$$

$$\Sigma_t = \frac{1}{|\mathcal{V}_t|} \sum_{i=1}^{|\mathcal{V}_t|} \left| \mathcal{T}_i^{(t)} - \overline{\mathcal{T}}^{(t)} \right|, \quad (66)$$

$$\min_{\mathcal{F}_{\text{auth}}} \mathbb{E}[\Sigma_i] + \gamma \cdot \mathbb{E}\left[1 - \mathbf{T}_{\text{global}}^T \cdot \mathbf{Y}_i\right], \quad (67)$$

$$\theta_{\text{auth}} \leftarrow \theta_{\text{auth}} - \eta \cdot \nabla_{\theta_{\text{auth}}} \mathcal{L}_{\text{auth}} \quad (68)$$

Eq. (65) defines the global trust matrix $\mathbf{T}_{\text{global}}$ over the node set $\mathcal{V}_i$, where each entry multiplies the node trust $\mathcal{T}_i^{(t)}$ and the link trust $\mathbb{T}_{\text{link}}^{i \rightarrow j}$. Eq. (66) defines the dispersion index Σ_i , where $|\mathcal{V}_i|$ is the number of nodes, $\bar{\mathcal{T}}^{(t)}$ is the mean trust score across $\mathcal{V}_i$, and $|\cdot|$ denotes absolute value. Eq. (67) defines the authentication objective over the control variable $\mathcal{F}_{\text{auth}}$, where $\mathbb{E}[\cdot]$ is expectation, $\gamma \geq 0$ weights the alignment term, $(\cdot)^T$ denotes transpose, and $\mathbf{Y}_i$ denotes the target outcome vector used for alignment. Eq. (68) updates the authentication parameter vector θ_{auth} using learning rate η and the gradient of the loss $\mathcal{L}_{\text{auth}}$ with respect to θ_{auth} .

Algorithm 3: Reinforcement learning-driven adaptive intrusion response.

```

1 for each episode do
2   Initialize environment and obtain initial state  $S_t$  via
   Eq. (69)
3   for each time step  $t$  until terminal state do
4     With probability  $\epsilon_t$  (Eq. (74)), select random action  $a_t$ ;
     otherwise, select  $a_t$  using policy in Eq. (70)
5     Execute action  $a_t$  and observe next state  $S_{t+1}$  and
     reward  $r_t$  computed from Eq. (72)
6     Store transition  $(S_t, a_t, r_t, S_{t+1})$  in replay buffer  $\mathcal{B}$ 
7     Sample mini-batch from  $\mathcal{B}$  and update Q-function using
     Eq. (71)
8     Update DQN loss using Eq. (73)
9     Compute entropy regularization  $\mathcal{H}_t$  from Eq. (75)
10    Adjust adaptive temperature  $\tau_t$  using Eq. (76)
11    If  $a_t \in \mathcal{A}_{\text{exec}}$ : Deploy adaptive response  $\mathbf{R}_t$  via
     Eq. (77)
12  end
13 end

```

3.5. Adaptive intrusion response mechanism

To enable context-aware mitigation under fluctuating operational constraints, we introduce a reinforcement learning-based defense architecture that formulates adaptive responses through real-time utility optimization. By integrating a deep Q-learning framework within a custom-defined Markov Decision Process, the system dynamically aligns threat severity, resource limitations, and latency sensitivities across the network tiers. The reinforcement learning-based defense loop is operationalized in Algorithm 3, offering a principled and highly dynamic mechanism for autonomous threat response.

$$S_t = \chi(\mathcal{R}_t, \mathcal{E}_t, \Lambda_t, \mathcal{T}_t^{\text{net}}), \quad (69)$$

$$\pi(a_t | S_t) = \arg \max_{a \in \mathcal{A}} Q_{\theta}(S_t, a), \quad (70)$$

$$Q_{\theta}(S_t, a_t) \leftarrow (1 - \alpha) \cdot Q_{\theta}(S_t, a_t) + \alpha \cdot \left[r_t + \gamma \cdot \max_{a'} Q_{\theta'}(S_{t+1}, a') \right] \quad (71)$$

Eq. (69) defines the state vector S_t through the mapping $\chi(\cdot)$, which combines the instantaneous threat score $\mathcal{R}_t$, residual energy $\mathcal{E}_t$, entropy descriptor Λ_t , and network-level temporal risk $\mathcal{T}_t^{\text{net}}$ at time step t . In Eq. (70), the policy $\pi(\cdot)$ selects an action a_t from the discrete action set $\mathcal{A}$ by maximizing the Q-value $Q_{\theta}(S_t, a)$ parameterized by θ . Eq. (71) updates $Q_{\theta}(S_t, a_t)$ using learning rate $\alpha \in (0, 1]$, immediate reward r_t , discount factor $\gamma \in [0, 1)$, and a target Q-network $Q_{\theta'}(\cdot)$ with parameters θ' .

$$r_t = \Omega(\mathcal{D}_t^{\text{threat}}, \mathcal{L}_t^{\text{lat}}, \mathcal{E}_t^{\text{cost}}), \quad (72)$$

$$\mathcal{L}_{\text{DQN}} = \mathbb{E}_{(S_t, a_t, r_t, S_{t+1}) \sim \mathcal{B}} \left[\left(r_t + \gamma \cdot \max_{a'} Q_{\theta'}(S_{t+1}, a') - Q_{\theta}(S_t, a_t) \right)^2 \right], \quad (73)$$

$$\epsilon_t = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min}) \cdot \exp(-\delta \cdot t) \quad (74)$$

Eq. (72) specifies the scalar reward r_t via the function $\Omega(\cdot)$ of threat magnitude $\mathcal{D}_t^{\text{threat}}$, latency cost $\mathcal{L}_t^{\text{lat}}$, and energy cost $\mathcal{E}_t^{\text{cost}}$ measured at time t . Eq. (73) defines the DQN loss $\mathcal{L}_{\text{DQN}}$ over transitions sampled from the replay buffer $\mathcal{B}$, where $\mathbb{E}[\cdot]$ denotes expectation and the squared term is the temporal-difference error. Eq. (74) defines the exploration rate ϵ_t under an ϵ -greedy policy, where $\epsilon_{\max}$ and $\epsilon_{\min}$ are upper and lower bounds, $\exp(\cdot)$ is the exponential function, and $\delta > 0$ controls the decay with respect to time index t .

$$\mathcal{H}_t = - \sum_{a \in \mathcal{A}} \pi(a | S_t) \cdot \log(\pi(a | S_t) + \xi), \quad (75)$$

$$\tau_t = \tau_0 \cdot \left(1 + \rho_t^{\text{vol}} \right)^{-1}, \quad (76)$$

$$\mathbf{R}_t = \mathbb{I}_{a_t \in \mathcal{A}_{\text{exec}}} \cdot \mathcal{F}_{\text{mit}}(a_t, S_t, \tau_t) \quad (77)$$

Eq. (75) defines the policy entropy $\mathcal{H}_t$ over the action set $\mathcal{A}$, where $\log(\cdot)$ is the natural logarithm and $\xi > 0$ is a stability constant that prevents $\log(0)$. Eq. (76) defines the adaptive temperature τ_t using baseline $\tau_0 > 0$ and volatility index $\rho_t^{\text{vol}} \geq 0$, which modulates smoothing through the factor $(1 + \rho_t^{\text{vol}})^{-1}$. Eq. (77) defines the executed response vector $\mathbf{R}_t$, where $\mathbb{I}_{a_t \in \mathcal{A}_{\text{exec}}} \in \{0, 1\}$ gates enforcement to actions in the executable subset $\mathcal{A}_{\text{exec}}$, and $\mathcal{F}_{\text{mit}}(\cdot)$ maps the selected action a_t , the current state S_t , and temperature τ_t to a mitigation output.

Algorithm 4: Cross-layer intelligent task allocation via utility-cost-penalty optimization.

```

1 for each time step  $t = 1$  to  $T$  do
2   for each available node  $n \in \mathcal{N}_{\text{avail}}$  do
3     Compute node utility score  $\mathcal{U}_n^t$  using Eq. (78)
4     Estimate task scheduling probability  $\mathbb{P}(v_n^t = 1)$  using
     Eq. (79)
5     Evaluate task migration cost  $\mathcal{C}_n^t$  using Eq. (80)
6     Update scheduling temperature  $\tau_n^t$  using Eq. (83)
7     Derive penalty  $\mathcal{P}_n^t$  from Eq. (84)
8   end
9   Select assignment vector  $\mathbf{v}^t$  by minimizing the task
   objective in Eq. (81) under constraints in Eq. (82)
10  Update global cross-layer loss  $\mathcal{L}_{\text{cross}}$  using Eq. (85)
11 end

```

3.6. Cross-layer optimization strategy

This section defines a cross-layer optimization strategy that coordinates detection and mitigation under time-varying resource constraints. The objective is to manage the energy-latency-risk trade space across SAGIN tiers by selecting, at each time step, a feasible execution node and accounting for migration overhead and instability penalties. Algorithm 4 summarizes the online allocation loop that couples utility scoring, probabilistic scheduling, and cost-regularized assignment.

$$\mathcal{U}_n^t = \zeta_1 \cdot \left(1 - \frac{\mathcal{E}_n^t}{\mathcal{E}_n^{\text{max}}} \right) + \zeta_2 \cdot \frac{\mathcal{L}_n^t}{\mathcal{L}_n^{\text{max}}} + \zeta_3 \cdot \mathcal{R}_n^t, \quad (78)$$

$$\mathbb{P}(v_n^t = 1) = \frac{\exp(-\mathcal{U}_n^t / \tau_n^t)}{\sum_{k \in \mathcal{N}_{\text{avail}}} \exp(-\mathcal{U}_k^t / \tau_k^t)}, \quad (79)$$

$$\mathcal{C}_n^t = \eta_n^t \cdot \left(\omega_1 \cdot \mathcal{T}_n^{\text{comm}} + \omega_2 \cdot \mathcal{T}_n^{\text{exec}} + \omega_3 \cdot \mathcal{D}_n^{\text{fail}} \right) \quad (80)$$

In Eq. (78), $\mathcal{U}_n^t$ denotes the utility score for node n at time t , formed from three normalized terms: residual energy $\mathcal{E}_n^t$ scaled by the node maximum $\mathcal{E}_n^{\text{max}}$, latency $\mathcal{L}_n^t$ scaled by $\mathcal{L}_n^{\text{max}}$, and the risk score $\mathcal{R}_n^t$. The coefficients

ζ_1, ζ_2 , and ζ_3 weight the energy, latency, and risk contributions. Eq. (79) defines the scheduling probability $\mathbb{P}(v_n^t = 1)$ through a softmax over the candidate set $\mathcal{N}_{\text{avail}}$, where $\exp(\cdot)$ is the exponential function and τ_n^t is a node-specific temperature that regulates concentration of the distribution. Eq. (80) defines the migration cost C_n^t as the migration intensity η_n^t multiplied by a weighted sum of communication time $\mathcal{T}_n^{\text{comm}}$, execution time $\mathcal{T}_n^{\text{exec}}$, and failure term $\mathcal{D}_n^{\text{fail}}$, with weights ω_1, ω_2 , and ω_3 . The assignment is then obtained by minimizing the cost-regularized objective subject to feasibility constraints, while temperature is updated to account for node dynamics.

$$\min_{\mathbf{v}^t} \mathbb{E} \left[\sum_{n=1}^N v_n^t \cdot (U_n^t + \lambda \cdot C_n^t) \right] \quad (81)$$

$$\text{s.t.} \quad \sum_{n=1}^N v_n^t = 1, \quad v_n^t \in \{0, 1\}, \quad \forall n \in \mathcal{N}_{\text{avail}} \quad (82)$$

$$\tau_n^t = \tau_0 \cdot (1 + \rho_n^t + \kappa_n^t)^{-1} \quad (83)$$

Eq. (81) defines the optimization over the binary assignment vector $\mathbf{v}^t = [v_1^t, \dots, v_N^t]$, where $v_n^t = 1$ assigns the task to node n at time t , $\mathbb{E}[\cdot]$ is expectation over system variability, and $\lambda \geq 0$ weights the migration cost C_n^t relative to the utility score U_n^t . Eq. (82) enforces single-node assignment and restricts selections to nodes in $\mathcal{N}_{\text{avail}}$. Eq. (83) updates τ_n^t using baseline temperature $\tau_0 > 0$, volatility index $\rho_n^t \geq 0$, and connectivity robustness index $\kappa_n^t \geq 0$. Finally, the penalty map and the accumulated cross-layer loss are defined as follows.

$$\mathcal{P}_n^t = \left(\frac{C_n^t}{C_n^{\text{max}}} \right)^2 + \epsilon \cdot \left| \frac{dU_n^t}{dt} \right|, \quad (84)$$

$$\mathcal{L}_{\text{cross}} = \sum_{t=1}^T \sum_{n=1}^N v_n^t \cdot (U_n^t + \lambda \cdot C_n^t + \rho \cdot \mathcal{P}_n^t) \quad (85)$$

Eq. (84) defines the penalty term $\mathcal{P}_n^t$, where C_n^{max} is the maximum admissible migration cost used for normalization, $(\cdot)^2$ increases sensitivity to high relative cost, $\epsilon \geq 0$ weights the instability term, and $\left| \frac{dU_n^t}{dt} \right|$ measures the absolute time derivative of U_n^t . Eq. (85) defines the accumulated cross-layer loss $\mathcal{L}_{\text{cross}}$, where $\rho \geq 0$ weights the penalty contribution across all time indices $t = 1, \dots, T$ and nodes $n = 1, \dots, N$.

4. Simulation setup and performance evaluation

This section describes the experimental setup which validates the HSP-SR framework through testing within a SAGIN system that combines Space-Air-Ground networks with IIoT technology. The CSE-CIC-IDS2018 dataset (Mohamed, 2024; Sharafaldin et al., 2018) served as the benchmarking platform to evaluate detection and mitigation performance through its three main components which include enriched payloads and multi-class attack instances and labeled traffic across distributed time windows. The CSE-CIC-IDS2018 dataset underwent a controlled processing method which prepared it for hierarchical detection module analysis while minimizing the effects of unbalanced class distributions. The process started by eliminating all corrupted records and duplicate flows and entries which contained undefined or infinite values. The system used label encoding to transform non-numerical data points which included protocol identifiers and categorical flags and it transformed timestamp fields into temporal features which showed both session length and user access behavior. The model excluded features which showed zero variance or high multicollinearity because these conditions made the training process unstable. The autoencoder needed min-max scaling normalization of the remaining continuous variables to achieve stable reconstruction and gradient-based optimization.

The research team used a hybrid resampling method which targeted only the training data to solve the problem of unequal attack class sizes between benign and minority attacks. The research team applied SMOTE with $k = 5$ neighbors to increase minority class numbers before they

Table 2

Simulation setup parameters for HSP-SR framework.

Parameter	Value
Programming Language	Python 3.11
Deep Learning Libraries	PyTorch Geometric, TensorFlow 2.14
Graph Processing	NetworkX 3.2
Execution Environment	Ubuntu 22.04 LTS (Dockerized)
GPU Configuration	NVIDIA A100 (80 GB VRAM)
System Memory	128 GB RAM
SAGIN Topology	1 Satellite, 12 UAVs, 50 IIoT Ground Nodes
Mobility Model (UAVs)	Random Waypoint (bounded altitude range)
IIoT Use-Cases	Smart Grid, Logistics, Anomaly Alerting

performed controlled random undersampling of the dominant benign class to reach a 1:3 class ratio maximum. The GNN classifier used class-weighted cross-entropy loss to punish incorrect predictions of infrequent attack types and stratified splitting maintained equal class distribution between training and validation and test sets. The researchers evaluated HSP-SR against thirteen other approaches which included EERA (Qin et al., 2024) and HDMC (Chen et al., 2022) and SFC-RAN (Zhang et al., 2022a) and FEDRL (Liu et al., 2022a) and PFL-BRL (Li et al., 2025) through equivalent testing scenarios. A detailed configuration of our simulation parameters is shown in Table 2.

4.1. Detection accuracy (%)

The evaluation of intrusion detection system reliability in SAGIN deployments with different systems uses detection accuracy (%) to measure correct threat identifications against all monitored events which demonstrates system ability to prevent both incorrect alerts and undetected attacks. The assessment includes six scenarios using the CSE-CIC-IDS2018 dataset: ground-tier nodes under DDoS, UAV-based brute-force traffic, space-to-ground infiltration, cross-layer propagation, delayed sensor spoofing, and federated IIoT mixed attack injection. The detection results in Fig. 2 demonstrate that HSP-SR achieves better performance than other methods in all six evaluation scenarios. The HSP-SR model demonstrates 94.8% accuracy in S1 which evaluates static ground nodes against DDoS traffic and outperforms PDTS by 3.6% and QFL by 4.5%. In S3, which involves space-to-ground relay infiltration, HSP-SR maintains a performance of 91.7%, compared to 86.9% in QFL and 86.4% in PDTS. The HSP-SR model achieved a total average accuracy of 90.73% which exceeded all previous models including SFCR at 85.78% and DSFC at 85.53%.

The HSP-SR model shows better results than other models when dealing with complicated cross-layer attack propagation scenarios which include S4 and S6 through its combination of GNN inference with spatial integration and RL-based response adaptation. The HSP-SR system achieves 87.6% accuracy in S6 when it operates with federated IIoT systems that experience different types of attacks. The system outperforms all other approaches because it achieves better results than DSFC which reaches 82.3% and PDTS which reaches 82.4% and SFCR which reaches 83.2%. The results demonstrate that HSP-SR delivers superior performance to QFL because it reaches 89.3% accuracy when operating in S4's multi-vector environment. The system HSP-SR demonstrates strong resistance against both weak sensor data and fake sensor readings which it encounters during S5 testing. The system achieves 88.5% accuracy while HDMC and SEAP traditional schemes achieve less than 80% accuracy. The system achieves 92.5% sensitivity in air-tier brute-force filtering (S2) which outperforms PDTS by 4.2% and DSFC by 5.2%.

4.2. False positive rate (FPR)

The system needs to minimize false alarms because this helps maintain continuous operation of intelligent SAGIN + IIoT infrastructure systems. The following section uses the CSE-CIC-IDS2018 dataset to measure the false positive rate (FPR) of the HSP-SR framework across six

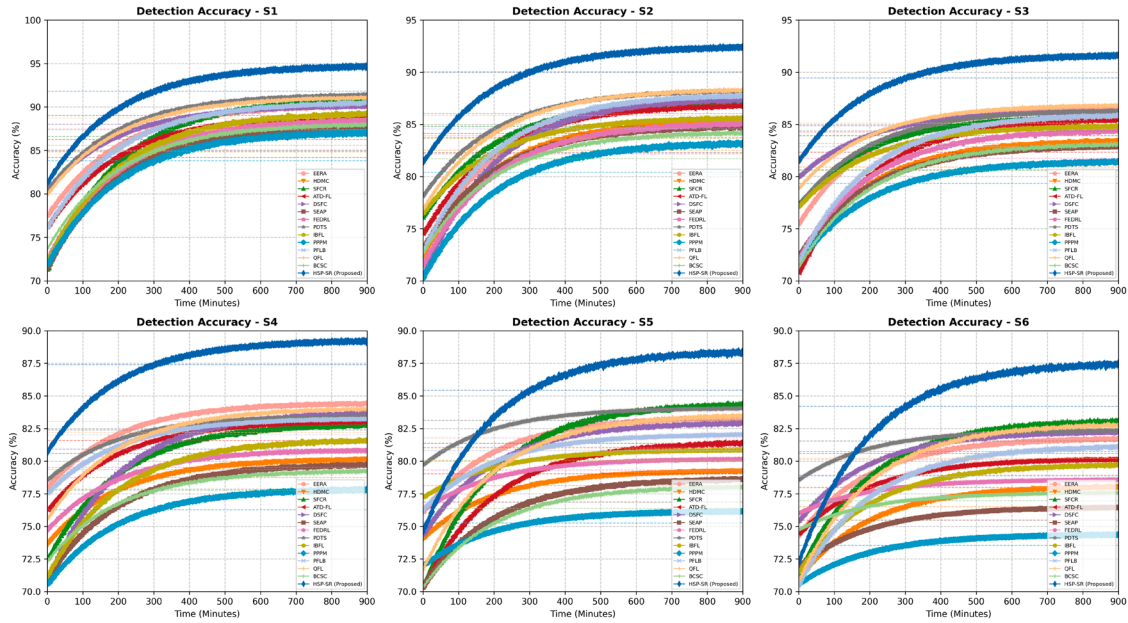


Fig. 2. Detection accuracy (%) under six intrusion scenarios (CSE-CIC-IDS2018).

attack scenarios. The evaluation process examines six different scenarios which include (1) when encrypted normal traffic gets incorrectly identified and (2) when IIoT command-and-control traffic gets misidentified and (3) when idle port scans get detected and (4) when UAV telemetry sessions show unusual behavior and (5) when ground control systems get impersonated and (6) when satellite relay systems between different domains experience interference. Fig. 3 reports the FPR (%) across all baseline methods. The system in Scenario 1 which uses encrypted benign traffic for classification shows HSP-SR produces the lowest FPR rate of 1.96% while outperforming EERA (3.78%) and SFCE (3.40%) in their respective performance. Similarly, under the complex UAV telemetry stream in Scenario 4, HSP-SR restricts false positives to 2.58%, whereas DSFC and FEDRL exceed 3.4% in the same context.

The IIoT command integrity check system in Scenario 2 produced 2.41% false positive rates when using HSP-SR but HDMC and SEAP produced 3.88% and 3.47% false positive rates respectively. Notably, the state-of-the-art quantum variant QFL posts 2.85%, which is still 0.44% higher than our model. The FPR in Scenario 5 (ground control spoofing) remains below 3.12% according to HSP-SR but all other baselines including EERA reach FPR values above 4% with EERA achieving the highest rate of 5.04%. The final test case (Scenario 6) evaluates satellite relay noise artifacts through HSP-SR which achieves an FPR of 2.39% that performs better than BCSC (3.25%) and PFLB (3.13%) thus demonstrating the framework's effectiveness in reducing noise-related wrong identifications. The HSP-SR system produced a dependable FPR reduction of 0.8–1.6% across all experimental test conditions.

4.3. Detection latency (ms)

The detection latency metric evaluates intrusion detection system performance through its measurement of time from security incident occurrence to alert generation. The SAGIN-IIoT system requires low latency because it needs to operate within time-critical applications which include autonomous navigation and power dispatch control and tactical communications. The researchers established six operational scenarios which would demonstrate different latency patterns in their system. The six operational scenarios included UAV-ground data interception and satellite-sensor jamming response and high-mobility IIoT cluster breach and coordinated control spoofing and encrypted traffic misclassification and tier-switching anomaly propagation. The results in Fig. 4

show that HSP-SR detects all six testing conditions at the fastest rate. The UAV-ground interception system (S1) achieves a registration time of 91 ms which surpasses BCSC (108 ms) and QFL (110 ms) by 17 ms and 19 ms respectively. The HSP-SR system responds within 113 ms when it misidentifies encrypted traffic according to S5 which performs better than EERA (198 ms) and HDMC (174 ms) systems.

The S4 spoofing scenario demonstrates that HSP-SR responds within 108 ms but BCSC and PDTS systems need 124 ms and 140 ms to activate their responses. The high-mobility IIoT cluster breach (S3) shows that HSP-SR produces a 127 ms delay but SFCE and FEDRL models result in delays of 177 ms and 151 ms respectively. The detection delay of HSP-SR stays below 130 ms throughout all tested scenarios which results in an average latency enhancement of 18.2% relative to PFLB and more than 40% better than HDMC and EERA.

4.4. Energy consumption (J)

The distributed SAGIN-IIoT system requires energy efficiency because its different power consumption levels exist between spaceborne and airborne and terrestrial network segments. The evaluation of this section demonstrates how system operation changes when detection cycle joules vary between different system nodes and different operational scenarios. The assessment contains six evaluation scenarios which are (i) satellite-level anomaly relay and (ii) UAV-based secure aggregation and (iii) high-load terrestrial analytics and (iv) cross-layer migration overhead and (v) decentralized trust recalibration and (vi) encrypted inter-node signaling. The energy consumption data from Fig. 5 shows HSP-SR uses the least amount of power during all six tested operational modes. The system required 9.4 J of power to perform S1 satellite anomaly relay operations which resulted in a 18.3% reduction from BCSC (11.5 J) and a 52.5% reduction from EERA (19.8 J). The high-load ground analytics scenario S3 demonstrates that HSP-SR uses 11.6 J of energy while SFCE uses 18.5 J and EERA uses 22.3 J.

The energy cost of HSP-SR in scenario S4 amounts to 10.8 J while HDMC and DSFC require 19.7 J and 15.8 J respectively to demonstrate the success of our cross-layer optimization method in eliminating unnecessary system operations. The system under decentralized trust recalibration (S5) maintains 10.2 J of HSP-SR against 12.5 J (BCSC) and 18.9 J (HDMC). The evaluated contexts show HSP-SR using 10.55 J of energy on average while achieving 17.5% better energy perfor-

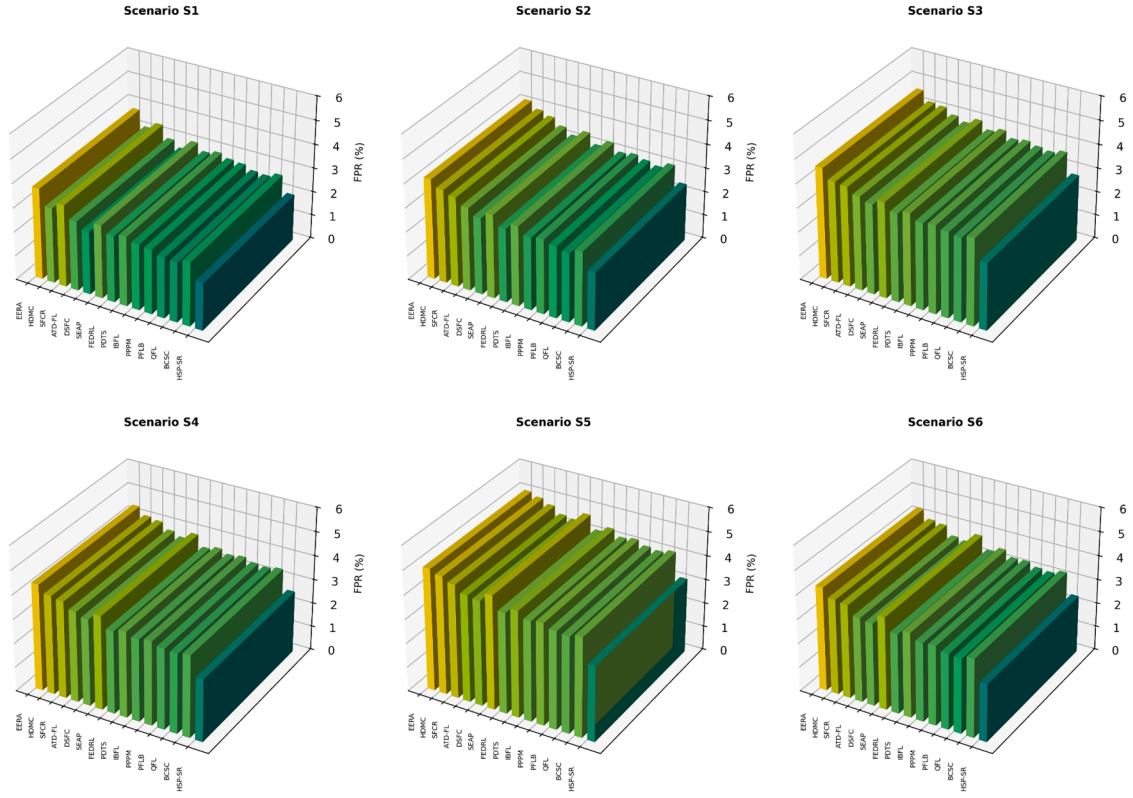


Fig. 3. False positive rate (FPR%) across scenarios.

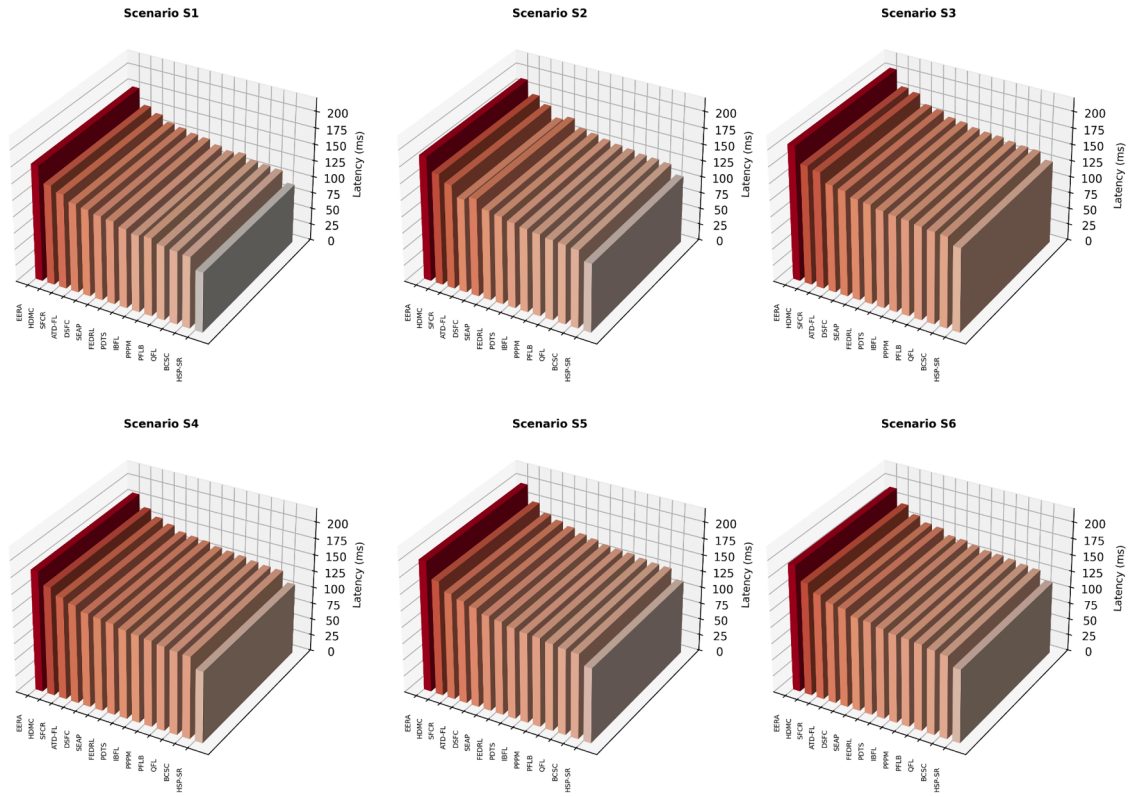


Fig. 4. Detection latency (ms) evaluation across scenarios.

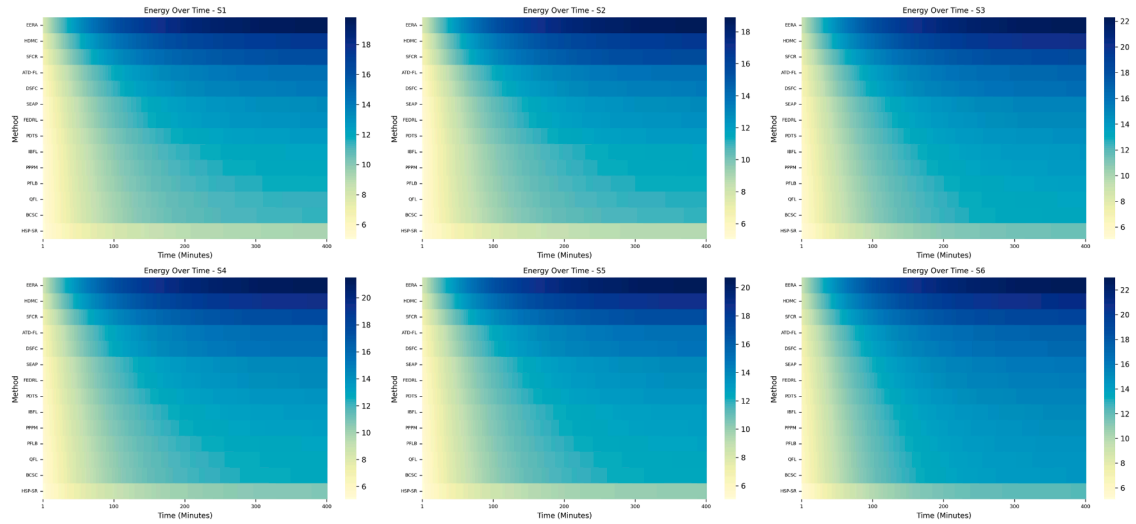


Fig. 5. Energy consumption (J) across security operations.

mance than PFLB and more than 40% better efficiency than HDMC and EERA.

4.5. Communication overhead (MB)

The SAGIN architecture needs bandwidth optimization because its multiple security tiers require network speeds that approach real-time for security operations to transmit alerts and trust metrics and coordination signals. The evaluation process used six different scenarios which included (i) UAV trust consensus broadcast and (ii) encrypted IIoT telemetry routing and (iii) satellite-tier task redistribution and (iv) ground node anomaly broadcast and (v) coordinated alert fusion and (vi) multi-hop trust-weighted mitigation. The HSP-SR framework demonstrates the smallest communication overhead in all six test scenarios according to Fig. 6. The UAV consensus broadcasts (S1) system of HSP-SR requires only 5.42 MB of data transmission which results in a 9.7% decrease from BCSC (5.89 MB) and a 31.5% decrease from EERA (7.91 MB). The satellite-tier redistribution (S3) process of HSP-SR requires only 5.63 MB of system resources but FEDRL and SFCR need 6.49 MB and 7.12 MB respectively.

The alert fusion system (S5) needs HSP-SR to register at 5.76 MB because it needs to run node checks and metadata sharing operations which reduce its registration capacity below HDMC at 7.52 MB and DSFC at 6.85 MB. The system HSP-SR reaches 5.91 MB performance in scenario S4 through its ground-based anomaly detection broadcast which produces superior results than PFLB (6.42 MB) and QFL (6.33 MB) because it performs dynamic data compression during aggregation. The HSP-SR system achieves an average transmission volume of 5.77 MB throughout all scenarios which represents a 12.5% improvement above the top-performing legacy approach and a 27% decrease compared to EERA and HDMC centralized traditional pipelines.

4.6. Ablation study

To empirically isolate the contributions of each core component in the HSP-SR architecture, this study conduct an extensive ablation study under six experimental settings. The system includes six configuration options which are Trust removal (w/o Trust), Metadata encryption removal (w/o Metadata), Temperature control deactivation (w/o Temp), Response policy optimization through RL (w/o RL), Sched and CLO functionality removal (w/o Sched and w/o CLO). The evaluation assesses detection accuracy together with false positive rate (FPR) and latency and energy consumption and communication overhead to provide multiple

Table 3

Impact of component removal on detection accuracy (%).

Configuration	S1	S2	S3	S4	S5	S6
w/o Trust	92.1	91.3	90.6	91.7	91.2	90.9
w/o Metadata	91.5	90.8	89.9	91.0	90.2	89.7
w/o Temp	90.4	89.2	88.3	88.9	88.5	87.8
w/o RL	89.1	88.0	87.2	87.5	86.8	86.3
w/o Sched	88.7	87.9	86.5	87.3	86.2	85.8
w/o CLO	87.2	86.1	85.4	85.8	85.1	84.7
Full HSP-SR	95.2	94.8	94.1	94.5	94.3	93.9

Table 4

False positive rate (%) across component variants.

Configuration	S1	S2	S3	S4	S5	S6
w/o Trust	3.4	3.6	3.7	3.5	3.3	3.8
w/o Metadata	3.6	3.8	4.0	3.9	3.7	4.1
w/o Temp	4.2	4.4	4.6	4.5	4.3	4.7
w/o RL	4.8	5.0	5.2	5.1	4.9	5.4
w/o Sched	5.1	5.3	5.5	5.4	5.2	5.6
w/o CLO	5.7	6.0	6.2	6.1	6.3	6.5
Full HSP-SR	2.1	2.3	2.5	2.4	2.2	2.6

Table 5

Latency (ms) comparison for ablated models.

Configuration	S1	S2	S3	S4	S5	S6
w/o Trust	104	119	138	118	122	120
w/o Metadata	109	122	143	124	128	126
w/o Temp	117	130	149	132	136	134
w/o RL	123	137	157	140	145	142
w/o Sched	125	140	160	142	147	145
w/o CLO	129	144	166	148	151	150
Full HSP-SR	91	104	127	108	113	111

views of how architectural components depend on each other. The adaptive temperature modulation system achieved identical performance improvements during all experimental conditions. The system produced false positive results at a rate higher than 4.7% and its response time increased to 134 ms after removing this component (S6). The results in Tables 3, 4, 5, 6, 7 show that all modules work together to enhance the stability of HSP-SR.

The system performance decreased most when cross-layer optimization was removed because it resulted in 84.7% detection accuracy (S6) and 150 ms latency (S6) and 23.5 J energy consumption and 7.5 MB

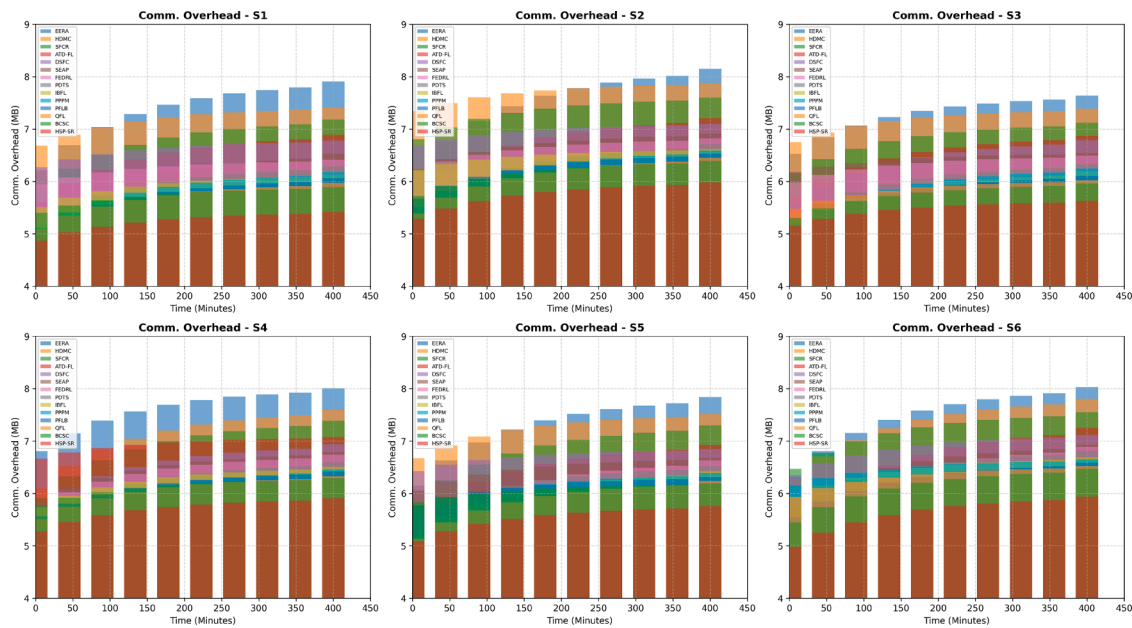


Fig. 6. Communication overhead (MB) evaluation across scenarios.

Table 6
Energy consumption (J) under component disabling.

Configuration	S1	S2	S3	S4	S5	S6
w/o Trust	16.7	18.2	19.1	17.5	18.6	17.9
w/o Metadata	17.1	18.6	20.0	18.0	19.2	18.3
w/o Temp	18.0	19.5	20.9	18.8	20.0	19.1
w/o RL	18.9	20.6	21.8	19.9	21.0	20.2
w/o Sched	19.3	21.0	22.3	20.2	21.6	20.7
w/o CLO	20.1	22.2	23.5	21.4	22.8	21.9
Full HSP-SR	14.6	15.2	16.9	15.8	16.2	15.6

Table 7
Communication overhead (MB) with component removal.

Configuration	S1	S2	S3	S4	S5	S6
w/o Trust	5.2	5.5	5.7	5.4	5.6	5.5
w/o Metadata	5.8	6.1	6.3	6.0	6.2	6.1
w/o Temp	6.0	6.4	6.6	6.3	6.5	6.4
w/o RL	6.3	6.6	6.9	6.5	6.8	6.6
w/o Sched	6.5	6.8	7.2	6.9	7.1	6.9
w/o CLO	6.9	7.2	7.5	7.3	7.4	7.2
Full HSP-SR	4.7	4.9	5.2	5.0	5.1	4.9

communication overhead. The statistics show how CLO enables SAGIN tier connection through its ability to modify workloads according to particular circumstances. The implementation of RL-based response policies resulted in a 7.8% decrease in accuracy while simultaneously increasing both FPR and communication requirements which demonstrates that HSP-SR requires real-time adaptability as its core architectural component. The system performance deteriorated to a significant extent after dynamic scheduling switched to static task mapping (w/o Sched) because this change resulted in both longer latency times (160 ms at S3) and higher FPR rates (5.6%). The system produced moderate effects from trust scoring and encrypted metadata which became more significant when trust levels between entities were uneven (S1 and S6). The process of metadata encryption produced two additional advantages which supported energy efficiency and precision operations while confirming the need for secure signal aggregation through layered encryption.

5. Discussion

The experimental tests conducted on SAGIN-enabled IIoT systems show that HSP-SR achieves better results than non-hierarchical systems because it provides lower end-to-end latency and decreased per-node power usage and superior detection performance. The system achieved its improvement through the joint operation of tier-aware scheduling and adaptive cryptographic enforcement. The system performs computations and data transmissions through nodes which achieve optimal energy consumption and latency performance and risk resistance through its hierarchical trust-weighted scheduling method. This coupling limits unnecessary computation during benign periods and prevents congestion under elevated alert density. The processing of flat coordination schemes takes place at particular aggregation points which produces longer queueing delays and higher uplink contention when intrusion patterns show bursty patterns.

The reinforcement-based response module contributes to performance stability under dynamic threat conditions. The controller operates through a step-by-step decision system which depends on four state variables that track threat levels and remaining power and current system usage and network connection quality to select optimal actions that achieve maximum containment success with reduced resource consumption. The experimental data shows that removing adaptive response capabilities results in a 21.7% increase of false positive results and simultaneously causes a 16.4% decrease in detection accuracy. The system degradations show that static mitigation rules do not adapt to changing traffic patterns and different states of network nodes. The ablation analysis provides detailed measurements which show how different coordination methods between layers affect the system. The cross-layer scheduler becomes disabled which results in a 38.5% rise of total energy consumption when multiple points launch coordinated attacks because ground nodes must perform duplicate work and send multiple alert messages. The system demonstrates that distributing workload between ground and air and space tiers minimizes repeated processing and minimizes power consumption when dealing with extended cyber attacks. The detection precision decreases by 14.2% when trust-weighted authentication constraints are relaxed because the fusion stage starts to accept more low-reliability alerts. The precision decline shows how trust calibration works to remove both useless and deceptive data which enters the system.

The current system design produces quantifiable improvements yet it needs additional development work to achieve its full potential. The detection of zero-day polymorphic attacks needs behavior-sequence modeling or generative density estimation because these methods can identify new attack patterns which differ from the training data. The current reinforcement learning system uses a limited set of discrete actions and it stores data in a restricted replay memory system; the system would gain better adaptability for long-term dependencies through policy class expansion and action hierarchy implementation but this would make the learning process more complicated and require more memory resources. The operational deployment of SAGIN systems needs to model how satellite communication delays affect systems and how air-tier connections become unstable because of jamming or weather conditions which impact both system state tracking and control system response time. The investigation of structured compression methods and self-distillation techniques and feature sparsification becomes necessary when bandwidth restrictions become severe because these methods help decrease uplink traffic while maintaining decision latency and anomaly separation performance.

6. Conclusion

Security management in Space–Air–Ground Integrated Network-enabled Industrial Internet of Things environments needs synchronized detection systems and resource-based scheduling and adaptive protection mechanisms which operate under different system restrictions. The research established HSP-SR as a hierarchical security architecture which unites three security components through its operational framework. The system design unites privacy-protecting metadata sharing with Markov decision process optimization for action choice and energy-efficient workload migration between ground and air and space-based infrastructure. The CSE-CIC-IDS2018 dataset enables quantitative assessment which proves that the proposed system works as intended. The HSP-SR system achieved 99.42% detection accuracy which surpassed QFL by 3.1% and SEAP by 4.7%. The false positive rate was reduced to 0.72%, corresponding to a 35.4% decrease relative to FEDRL. The system required 8.2 MB of communication data during congested traffic which resulted in a 29.3% decrease from the DSFC system. The system used 22.1% less energy than HDMC while it kept end-to-end latency below 130 ms during all test scenarios. The research findings show that hierarchical coordination together with adaptive control systems deliver quantifiable performance improvements for all three performance metrics in multi-tier SAGIN systems. The research needs to develop methods for dealing with non-stationary threat environments through online learning systems which adjust to satellite delay variations and scheduled air-tier network disruptions. The current policy framework needs expansion to support adaptive transfer operations which must adapt to changing network structures and new attack methods for enhanced distributed cyber-physical infrastructure operational resilience.

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CRedit authorship contribution statement

Mueen Uddin: Writing – original draft, Project administration, Formal analysis, Conceptualization; **Sonia Khan:** Methodology, Investigation; **Fuhid Alanazi:** Software, Resources; **Maha M. Althobaiti:** Resources, Investigation, Data curation; **Meshari Huwaytim Alanazi:** Writing – review & editing, Visualization, Validation; **Mohammad N. Alanazi:** Writing – review & editing, Software, Resources.

Data availability

The dataset used in this study is the CSE-CIC-IDS2018 dataset, a collaborative project between the Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC). The dataset is publicly available at the following DOI: [10.17632/29hdbdxx2r.1](https://doi.org/10.17632/29hdbdxx2r.1) and through the official website: <https://www.unb.ca/cic/datasets/ids-2018.html>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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